

A Contragredient View of Conformal Placement and Steinitz Balancing

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July 2026

ROUGH DRAFT — NOT FOR DISSEMINATION

Abstract

We give the validity of split conformal prediction an exact contragredient form: coverage is an orbit average, and the Reynolds identity $\langle P^\vee f, v \rangle = \langle f, Pv \rangle$ lets it be read either as transporting the evaluation functional or as averaging the data. The primal construction under the same permutation action is Steinitz balancing: ordering a zero-sum population so that every prefix sum stays bounded. Conformal placements form such a population, and we construct an explicit ordering whose running coverage stays within $1/(2t)$ of nominal, exact on the full orbit. On the function–measure pairing, the literal transpose of calibration is balanced sampling and exact cubature. Exchangeability is untouched: it alone makes the designated test placement an even draw.

1 Introduction

Split conformal prediction validates its coverage claim by a symmetry argument over sample slots. This note gives that argument an exact dual form and identifies the deterministic construction that lives beside it under the same group action. The identity behind conformal validity is a contragredient statement for the permutation action on slots (Section 3). The primal construction under that action is *Steinitz balancing*: ordering a finite zero-sum population so that every prefix sum stays uniformly bounded (Steinitz, 1913). Conformal placements are themselves a scalar Steinitz population (Section 5), and admit a sharper balanced ordering than the general lemma provides. On the function–measure pairing, the literal transpose of calibration is balanced sampling and exact cubature (Section 6). Section 7 records how the same two-ingredient structure appears in the dependent-data floor of Barber and Pananjady (2026), and Section 8 separates three rates that the shared algebra can blur. As far as we know, the connection between conformal placement and balanced orderings has not been drawn before. Herding (Welling, 2009), whose $O(1/n)$ bookkeeping resembles the constructions here, is the with-replacement, continuous-domain relaxation of the finite balanced-permutation problem (Harvey and Samadi, 2014); it appears only in Remark 2.

2 Conformal calibration and placement

Fix $\alpha \in (0, 1)$, write $p = 1 - \alpha$, let $S_1, \dots, S_n$ be calibration scores and S_{n+1} a test score, almost surely distinct, and let $k = \lceil p(n+1) \rceil \leq n$. Write $N = n + 1$. Split conformal declares coverage when $S_{n+1} \leq q_n$, with $q_n = S_{(k)}$ the k -th smallest calibration score.

Lemma 1 (Steering identity). *With $q_n = S_{(k)}$, $h_q(s) = \mathbf{1}\{s \leq q\}$, and the scores distinct,*

$$\sum_{i=1}^n (h_{q_n}(S_i) - p) = k - pn \in [p, 1 + p).$$

Proof. Exactly k calibration scores satisfy $S_i \leq S_{(k)}$; and $p(n+1) \leq k < p(n+1) + 1$. $\square$

The material of this section is standard (Papadopoulos et al., 2002; Vovk et al., 2005; Lei et al., 2018).

Proposition 1 (Counting identity). *Let $s_1, \dots, s_N$ be distinct reals and, for $i \in \{1, \dots, N\}$, call placement i covered if s_i is at most the k -th smallest of $\{s_j\}_{j \neq i}$, writing $\chi_i(s) = \mathbf{1}\{s_i \leq Q_k(s_{-i})\}$ for the indicator. Then $\chi_i(s) = 1$ iff the rank of s_i among all N values is at most k , and consequently $\sum_{i=1}^N \chi_i(s) = k$.*

Proof. Placement i is covered iff at most $k-1$ of the other values are smaller than s_i , iff its overall rank is at most k ; ranks are a permutation of $\{1, \dots, N\}$. $\square$

Corollary 1 (Marginal validity). *If $(S_1, \dots, S_N)$ is exchangeable with almost surely distinct components, then $\mathbb{P}\{S_N \leq q_n\} = k/N \geq 1 - \alpha$.*

Proof. The coverage event is the event that placement N is covered. By exchangeability every placement indicator has the same expectation; average Proposition 1. $\square$

The control-theoretic reading of the calibration step is established in the online literature: the level update of adaptive conformal inference (Gibbs and Candès, 2021) is integral feedback on realized miscoverage, and Angelopoulos et al. (2023) name the identification. There the controlled quantity is a time average along a sequence; Lemma 1 is the batch counterpart, a single dead-beat correction.

3 The contragredient identity

Let $V = E^N$ for a vector space E , with $\mathfrak{S}_N$ acting by permuting slots, $(\rho(\pi)v)_j = v_{\pi^{-1}(j)}$, and let $\rho^\vee(\pi)f = f \circ \rho(\pi^{-1})$ be the contragredient action on functionals (for the dual representation and the averaging operator see, e.g., Serre, 1977). For the prefix-sum functionals $L_t(v) = \sum_{j=1}^t v_j$ one has, exactly,

$$L_t(\rho(\pi)v) = (\rho^\vee(\pi^{-1})L_t)(v) : \quad (1)$$

permute the points holding the prefix functional fixed, or hold the points fixed and permute the functional. Averaging over the group, let $P = \frac{1}{N!} \sum_{\pi} \rho(\pi)$ be the Reynolds projector and $P^\vee = \frac{1}{N!} \sum_{\pi} \rho^\vee(\pi)$ its dual, so that

$$\langle P^\vee f, v \rangle = \langle f, Pv \rangle \quad \text{for all } f \in V^*, v \in V. \quad (2)$$

For the permutation representation, $Pv = \bar{v} \mathbf{1}$ and $P^\vee e_N^* = \frac{1}{N} \mathbf{1}^*$.

This is the formal statement behind the two descriptions of conformal validity. Define the centered placement residual vector $r(s) = (\chi_i(s) - k/N)_{i \leq N}$; Proposition 1 says $\langle \mathbf{1}^*, r(s) \rangle = 0$, i.e.

$$\langle P^\vee e_N^*, r(s) \rangle = 0 \quad \text{for every distinct } s. \quad (3)$$

Exchangeability then converts the orbit average into the law of the designated placement: $\mathbb{E} \chi_N(S) = \mathbb{E} \langle P^\vee e_N^*, \chi(S) \rangle = k/N$, which is Corollary 1. The identity is exact and requires no analogy; it is the defining identity of the dual representation, applied to the acceptance vector. What it does not do is remove exchangeability: (3) is a statement about the whole orbit, and only exchangeability makes the one designated test placement uniform on it.

4 Steinitz balancing

The primal construction living under the same $\mathfrak{S}_N$ action is the ordering of a finite zero-sum population. If $z_1, \dots, z_N \in E$ satisfy $\sum_i z_i = 0$ and $\|z_i\| \leq 1$, the Steinitz lemma (Steinitz, 1913) supplies a permutation π with every prefix bounded, $\|\sum_{j \leq t} z_{\pi(j)}\| \leq C_E$ for all t , with C_E depending on the space and norm but not on N (a dimension-linear bound holds for arbitrary norms; Grinberg and Sevast'yanov, 1980); hence running averages within C_E/t of zero. This is the bounded-ledger $O(1/t)$ phenomenon, obtained by an element of the same permutation group that Section 3 acts with, and its dual operation really is the transport of a prefix functional through (1).

For the relation to herding, see Remark 2.

5 Balanced conformal placements

Conformal placements are themselves a scalar Steinitz population: by Proposition 1 the residuals $r_i(s) = \chi_i(s) - k/N$ satisfy $\sum_i r_i(s) = 0$ and $|r_i(s)| \leq 1$. Because they take only two values, the ordering can be made sharper than the general lemma provides.

Proposition 2 (Balanced placement ordering). *For every distinct score vector s there is a permutation $\pi \in \mathfrak{S}_N$ such that, simultaneously for every $t = 1, \dots, N$,*

$$\left| \sum_{j=1}^t \chi_{\pi(j)}(s) - \frac{tk}{N} \right| \leq \frac{1}{2}, \quad \text{hence} \quad \left| \frac{1}{t} \sum_{j=1}^t \chi_{\pi(j)}(s) - \frac{k}{N} \right| \leq \frac{1}{2t},$$

with equality to zero at $t = N$.

Proof. Define the balanced binary word $b_t = \lfloor tk/N + \frac{1}{2} \rfloor - \lfloor (t-1)k/N + \frac{1}{2} \rfloor$. Since $0 \leq k/N \leq 1$, each $b_t \in \{0, 1\}$, and $\sum_{j \leq t} b_j = \lfloor tk/N + \frac{1}{2} \rfloor$, so $|\sum_{j \leq t} b_j - tk/N| \leq \frac{1}{2}$ for every t , with exactly k ones among $b_1, \dots, b_N$. Assign the k covered placement indices to the positions with $b_t = 1$ and the uncovered indices to the rest. $\square$

This is one-dimensional balanced rounding, performed on the conformal placement orbit itself; prefix bounds of this type are classical, going back to the chairman assignment problem (Tijdeman, 1980). In the notation of Section 3 the two readings are literal: $\langle L_t, \rho(\pi)r(s) \rangle = \langle \rho^\vee(\pi^{-1})L_t, r(s) \rangle$ — move the residual vector through the permutation holding the prefix test fixed, or hold the residuals fixed and move the nested family of prefix functionals.

The proposition gives a deterministic traversal of the N possible evaluation placements whose running coverage is calibrated to $O(1/t)$, which is what orbit computations and repeated-placement audits need. It does not replace exchangeability in one-shot conformal prediction, where a single placement is designated as the test; only exchangeability makes that placement uniform on the orbit.

6 Balanced sampling: the function–measure transpose

If the pairing of interest is $\langle h, \nu \rangle = \int h d\nu$ rather than the permutation representation, the literal transpose of calibration is balanced sampling. Calibration solves: given the empirical score measure ν_n , find a threshold test h_q with $\langle h_q, \nu_n \rangle = k/n$. Its transpose solves: given tests $h_1, \dots, h_d$ and targets $m_1, \dots, m_d$, find a measure ν with $\langle h_j, \nu \rangle = m_j$. Over a finite candidate population with $A_{ji} = h_j(x_i)$ this is the selection problem $Az = b$, $z \in \{0, 1\}^N$, which is precisely the balanced-sampling problem solved (exactly when geometry and integrality permit, approximately otherwise)

by the cube method (Deville and Tillé, 2004). For weighted atomic measures, convex geometry does better: $\mu = \int \phi dP$ lies in the convex hull of $\phi(\mathcal{X})$, so Carathéodory supplies at most $d + 1$ weighted points matching all d moments exactly, the Tchakaloff side of exact cubature (Tchakaloff, 1957). In this formulation, balanced sampling and exact cubature are the dual problem.

A designed calibration set sits inside this frame. If designed scores $S_1, \dots, S_n$ have empirical distribution function within Kolmogorov distance D of a continuous reference law $\bar{F}$, and the test score is drawn from $\bar{F}$ independently of the design, then $|\bar{F}(S_{(k)}) - k/(n+1)| \leq D + 1/(n+1)$ and $|\bar{F}(S_{(k)}) - p| < D + (1+p)/n$, deterministically in the design (the proof is $\hat{F}_n(S_{(k)}) = k/n$ plus Lemma 1). A quantile grid achieves $D \leq 1/(2n)$. This is design-based rather than distribution-free: it suppresses the $n^{-1/2}$ fluctuation of conditional coverage at the price of assuming the target law, and whether reproducing-kernel guarantees of kernel herding or kernel thinning (Chen et al., 2010; Dwivedi and Mackey, 2021) control the Kolmogorov discrepancy required here is left open.

7 The Barber–Pananjady floor

The two-ingredient structure of the preceding sections reappears in the dependent-data coverage floor of Barber and Pananjady (2026). For $S = (S_1, \dots, S_N)$ they define subvectors $\Delta_{k,\tau}^0(S)$ and $\Delta_{k,\tau}^1(S)$ of length $N - \tau$, deleting a block of τ entries while keeping S_N (respectively rotating S_k) in the final slot, and set

$$\Psi_{k,\tau}(S) = d_{\text{TV}}(\mathcal{L}(\Delta_{k,\tau}^0(S)), \mathcal{L}(\Delta_{k,\tau}^1(S))), \quad \bar{\Psi}_\tau = \frac{1}{N} \sum_{k=1}^N \Psi_{k,\tau}(S),$$

with $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)|$. Their Theorem 5 gives, for a pretrained memoryless score,

$$\mathbb{P}\{S_N \leq q_n\} \geq 1 - \alpha - \min_{0 \leq \tau \leq n} \left\{ \frac{\tau}{N} + \bar{\Psi}_\tau \right\},$$

their Corollary 6 bounds $\bar{\Psi}_\tau \leq 2\beta(\tau)$ under stationary β -mixing, their Theorem 8 supplies a matching ceiling, and their Theorem 7 shows tightness up to the factor $(1 - \alpha)/4$.

The argument separates into an exact insertion–deletion stability statement for empirical quantile *levels* (deleting τ entries moves the rank of a fixed threshold by at most τ , so the level shifts by at most τ/N : order-statistic geometry, controlling levels rather than values, since empirical quantile values are not stable under small total-variation perturbations) and a total-variation comparison between switched laws, which is the definition of the switch coefficient. Both ingredients are linear in their perturbation sizes and no concentration step appears; a blocking argument (Oliveira et al., 2024) bounds the same quantity through a maximal inequality and pays $\sqrt{\tau/n}$.

8 Three rates

The shared algebra can blur three quantities. The marginal coverage k/N of Corollary 1 is an exact symmetry identity; its $O(1/n)$ deviation from p is rank-grid rounding, not estimation error. The conditional coverage $\bar{F}(q_n)$ of a randomly sampled calibration set fluctuates around nominal at the sampling scale $n^{-1/2}$; the design of Section 6 suppresses this at the price of assuming $\bar{F}$. Steinitz discrepancies are genuine approximation errors, $O(1/t)$ with constants from the Steinitz lemma. The ledgers match; the objects differ.

9 Remarks

Remark 1 (The pinball dual). Threshold calibration also has an exact convex dual. The empirical p -quantile minimises the pinball loss (Koenker and Bassett, 1978), $\sum_i \rho_p(S_i - q)$ with $\rho_p(r) = \sup_{u \in [p-1, p]} ur$, and Fenchel duality gives

$$\max \left\{ \sum_i u_i S_i : u_i \in [p-1, p], \sum_i u_i = 0 \right\},$$

with $u_i = p - h_q(S_i)$ away from the threshold tie. The dual optimum is a balanced signed measure on the scores: calibration’s exact optimization dual, though not itself a point-generation algorithm.

Remark 2 (Herding). Herding (Welling, 2009) generates points by the recursion $w_t = w_{t-1} + \mu - \phi(x_t)$, $x_t = \arg \max_x \langle w_{t-1}, \phi(x) \rangle$, and attains $O(1/n)$ feature-mean discrepancy conditional on bounded weights; the identification with Frank–Wolfe (Bach et al., 2012) shows the fast rate needs interiority-type conditions, with Chen et al. (2010) treating the kernel case. Harvey and Samadi (2014) pose herding over a finite population as exactly the balanced-permutation problem of Section 4, making generic herding the with-replacement relaxation of Steinitz balancing; on the pairing of Section 6 it is one iterative approximate solver for balanced sampling. The quadrature triangle is older on that side: optimally-weighted herding is Bayesian quadrature (Huszár and Duvenaud, 2012), and Snell and Griffiths (2025) recently recast the conformal computation as Bayesian quadrature over the calibration scores.

Remark 3 (Limits). The passage from orbit identity to one-shot coverage is exchangeability’s, or its mixing surrogate’s in Section 7; no bookkeeping supplies it. Nor does control of thresholds and labels create information: it re-levels coverage while conditional sharpness remains a property of the scores (Cotton). Whether placement balancing can be aimed conditionally, selecting or reweighting calibration slots to balance the rank count given the input, is open; localized and weighted conformal methods are the nearest existing steps.

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